A pvlib-Based Computational Model of a Multi-Inverter Rooftop Photovoltaic Plant Driven by Public Weather Data: Sensitivity Analysis and Field Validation

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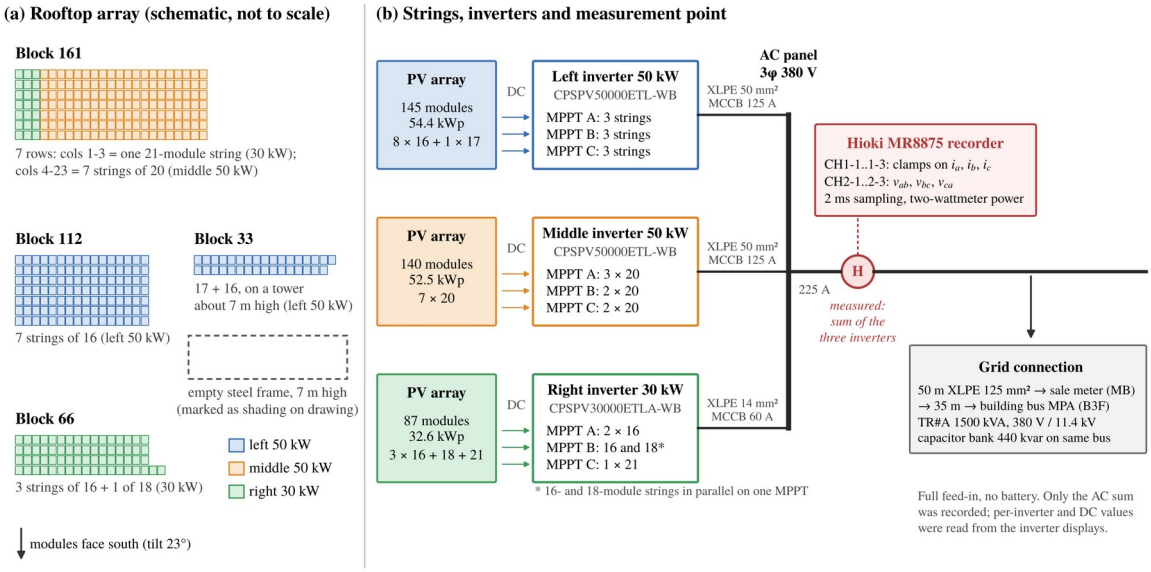


Fig. 1 Plant layout. (a) Roof blocks and their inverter (positions schematic). (b) Strings per MPPT, AC connection and the recorder at the common AC output of the three inverters.

Abstract

Rooftop photovoltaic (PV) energy output is frequently estimated using datasheets and publicly available weather data. In this study, a 139.5 kWp PV plant in Taiwan (comprising 372 modules, one 30 kW inverter and two 50 kW inverters) is modeled using pvlib. Nine uncertain factors are ranked through a converged Sobol analysis (45,056 runs), and the model is compared with AC power measurements at the common inverter output as well as with the lifetime energy counters of the inverters. DC loss and irradiation error account for 84% of the annual variance, while the inverter model contributes 7%. The inverter effect is primarily due to the assumed efficiency (2.6%) rather than the model structure (0.4–0.6%). On clear autumn and winter days, the plant delivered 80–89% of the modeled energy (normalized mean bias error up to +24%), although the hourly profile is accurately reproduced (correlation coefficient r ≥ 0.97). The uncertainty budget only matches the observations when all of the factors are at their minimum. Over a 3.5-year period, the inverter counters agree with the model to within 3%. The model is therefore suitable for annual energy estimates but biased at the hourly level on clear days, and on-site irradiance measurements are needed to identify the root causes.

Keywords: Photovoltaic; Inverter modeling; pvlib; Uncertainty budget; Field validation

1. Introduction

The energy of a PV plant can be estimated from module and inverter datasheets and public weather data with pvlib [1], [2], but the result depends on the assumption made at each step. For the inverter, a constant efficiency is the simplest choice, PVWatts [3] adds a load-dependent efficiency, and the Sandia [4] and ADR [5] models need test data that datasheets rarely give; hourly data also underestimate clipping [6]. Uncertainty studies point to plane-of-array modeling [7], 2–6% annual uncertainty of satellite irradiation [8] and derate assumptions [9], but seldom treat a single rooftop plant whose weather data come from a distant station.

For the 139.5 kWp rooftop plant of National Changhua University of Education (NCUE), Taiwan, we ask: (i) how large the effect of the inverter model is compared with other factors, and whether it stems from model structure or assumed efficiency; (ii) how much the weather source (a station 21 km away or ERA5 [10]) matters; and (iii) whether the variance-based uncertainty budget [11], [12] encloses the energy the plant delivers.

2. Plant, Data and Model

2.1 Plant

The plant (24.066°N, 120.559°E) has 372 AJP-S672-375 modules [13] on open racks, facing south at 23° tilt, and feeds one 30 kW and two 50 kW inverters [14] without storage (Fig. 1, Table 1). A site inspection in October 2026, based on the hand annotations of the drawings, the module count of each roof block and the string currents and MPPT voltages on the inverter displays, found a layout that differs from the drawings (144/144/84 modules): 145, 140 and 87 modules in 21 strings of 16 to 21 modules (Fig. 1b). For the annual energy this changes the result by only +0.02%. Two blocks of the left inverter stand on a 7 m tower next to an empty frame of the same height, the middle inverter logs fan faults with de-rating, and the counter of the 30 kW inverter is faulty.

Table 1 Plant data (as built).

Item	Value
Module	375 W, γ = −0.44 %/°C, NOCT 46 ± 2 °C
Inverter 30 / 50 kW	η_{max} 98.6 / 98.5%
Modules (left / middle / right)	145 / 140 / 87
Capacity (kWp)	54.4 / 52.5 / 32.6
DC/AC ratio	1.09 / 1.05 / 1.09

2.2 Weather data

Hourly irradiation and air temperature come from the CWA Tianzhong station (467270) [15], 21.4 km from the site (1626 kWh/m² in 2023), and hourly ERA5 data from PVGIS [16], [17]. The hour label refers to the end of the hour (1 MJ/m² per hour = 277.8 W/m²), and one missing hour was interpolated. No irradiance sensor exists on the roof.

2.3 Power measurement

A Hioki MR8875 [18] records the three line currents and line-to-line voltages at the common AC output (Fig. 1b). The bus voltages are three sinusoids displaced by 120° (Fig. 2); for balanced conditions the active power is

$$P = \sqrt{3}\, V_{LL}\, I_L \cos\varphi \qquad (1)$$

With three wires and no neutral, the instantaneous power is obtained from two products (two-wattmeter method), which also applies under unbalance and distortion [28]:

$$P = \langle v_{ab}\, i_a - v_{bc}\, i_c \rangle \qquad (2)$$

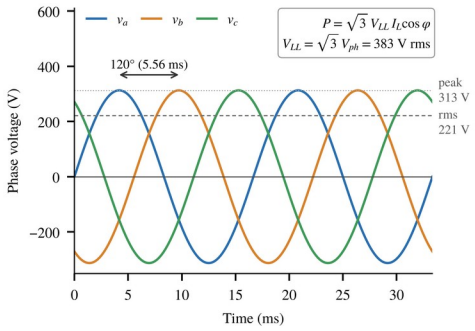


Fig. 2 Phase voltages of the AC bus (60 Hz): equal amplitude, displaced by 120°; 221 V rms per phase, 383 V rms line-to-line.

Because $v_{ab} = v_a - v_b$, a line-to-line voltage is √3 times the phase voltage and 30° ahead of it. Multiplying each current by its corresponding line-to-line voltage would therefore give 3 cos 30°/√3 = 1.5 times the true power. The voltages are 120° ± 1.5° apart, the currents lag by 0.5° to 4.3°, the unbalance is less than 3.8% and the power factor is 0.995. Eq. (1) applied to the fundamentals agrees with Eq. (2) to within 0.13% (median).

The main dataset (collected 16 to 19 October 2026) was sampled at 2 ms, thinned to 10 ms and averaged to give 4,201 one-minute values; no data gap was longer than 90 s. Seven further two-hour datasets (from 2025 to 2026) were thinned to 20 ms. Thinning without filtering introduces aliasing of the 60 Hz waveforms; however, at 10 ms and 20 ms the samples are distributed across five equally spaced points in the cycle, allowing the power ripple to average out over one minute. Sampling intervals corresponding to whole cycles, such as 200 ms, were avoided because they sample a single point in the cycle. An evaluation using 400 synthetic minutes (incorporating the measured unbalance, harmonics and ±30% power variation) confirmed this: the maximum error of the one-minute power was 0.06% for 10 ms and 0.05% for 20 ms intervals, and 0.12% when the grid frequency was exactly 60 Hz, whereas errors reached 12% for 200 ms intervals.

The measurement chain was not calibrated. The night-time offset is −0.011 kW, and on 18 October the inverter displays showed 3.0% more energy than the recorder (572 against 555 kWh at 16:58); the middle inverter showed 39.3% of the recorded energy, against a module share of 37.6%.

2.4 Model chain

Fig. 3 illustrates the sequence comprising Erbs decomposition [19], Perez transposition [20] with an albedo of 0.25, Ross cell temperature [21] and PVWatts DC power [3]:

$$P_{dc} = P_0(G/1000)[1 + \gamma(T_c - 25)](1 - f_{dc}) \qquad (3)$$

Here, f_{dc} is set to 8.7% (or 11.4% when the default shading loss is applied). Each inverter is modeled in three ways: constant efficiency, PVWatts-DS (datasheet peak η as nominal, representing an upper bound) or PVWatts-G (η 0.96, the pvlib default); our implementation matches

pvlib.inverter.pvwatts to within 0 W. A model variant incorporates the Martin–Ruiz incidence-angle loss [22], and the performance ratio is determined according to IEC 61724-1 [23].

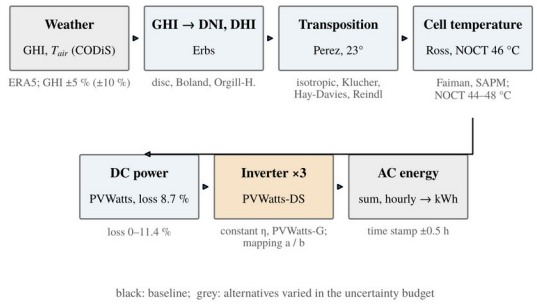


Fig. 3 Model chain. Grey: alternatives varied in Section 3 (e.g. Faiman [24] and SAPM [25] cell temperature).

3. Annual Uncertainty Budget

Nine factors were varied with SALib [26] (Table 2): continuous factors uniformly over the stated ranges, model options with equal weight. The ranges are assumptions with the given basis, and the ranking depends on them, as the ±5% and ±10% GHI cases show.

Table 2 Factors and ranges (model options with equal weight) and Sobol total index S_T for ±5% and ±10% GHI error (4,096 base samples, 95% interval).

Factor	Range, distribution, basis	S_T ±5%	S_T ±10%
DC loss	0–11.4%, uniform; PVWatts loss without / with shading [3]	0.495 ± 0.020	0.243 ± 0.010
GHI error	±5% (±10%), uniform; sensor class, 21 km distance [8]	0.344 ± 0.014	0.678 ± 0.025
Inverter	constant η, PVWatts-DS, PVWatts-G	0.071 ± 0.003	0.035 ± 0.001
Transpos.	Perez, isotropic, Klucher, Hay-Davies, Reindl	0.058 ± 0.003	0.028 ± 0.001
Cell temp.	Ross, Faiman, SAPM	0.029 ± 0.001	0.015 ± 0.001
GHI split	Erbs, disc, Boland, Orgill-Hollands	0.003 ± 0.000	0.001 ± 0.000
NOCT	44–48 °C, uniform; datasheet 46 ± 2	0.002 ± 0.000	0.001 ± 0.000
Time shift	−0.5, 0, +0.5 h (hour label)	0.001 ± 0.000	0.000 ± 0.000
Mapping	two layouts within inverter limits	0.000 ± 0.000	0.000 ± 0.000

3.1 Results and convergence

The baseline gives 195 MWh AC (1399 kWh/kWp), with the 5th to 95th percentile ranging from −5.5% to +11.4%. DC loss (−3.0% to +9.6%) and GHI error (±5%) account for 84% of the variance, and the inverter model ranks third (Fig. 4; 3% with ±10% GHI error). The total indices vary by no more than 0.004 between 1,024 and 4,096 base samples, and their standard deviation across six scrambling seeds is less than 0.002. The ranking is stable from 64 samples, and the intervals for the first five factors do not overlap. Interactions are negligible ($\Sigma S_T - \Sigma S_1 = 0.005$), the layout mapping changes the energy by less than 0.01%, and no inverter clips at the baseline loss; without DC loss, clipping is limited to a maximum of 21 kWh per year.

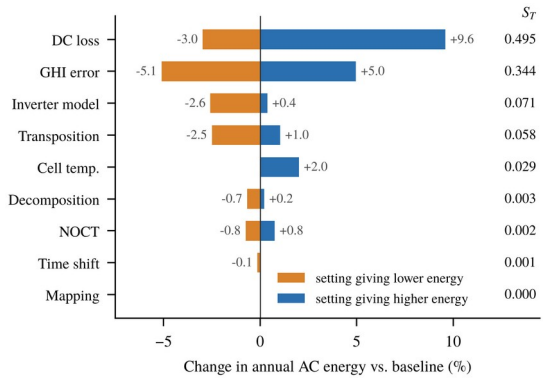


Fig. 4 Change in annual AC energy when one factor moves within its range (orange: setting giving the lowest, blue: the highest energy) and Sobol total index S_T (4,096 base samples).

3.2 Inverter: structure or efficiency

The formulations differ in curve shape and in assumed efficiency at once. Fig. 5 separates the two: at equal η, constant efficiency and PVWatts differ by 0.36–0.63%, whereas lowering η from 0.985 to 0.96 removes 2.6%. The inverter share of the budget is therefore an efficiency assumption, not a structural effect. On the clear October days, when most energy is produced near full load, the two structures with equal η differ by less than 0.1%.

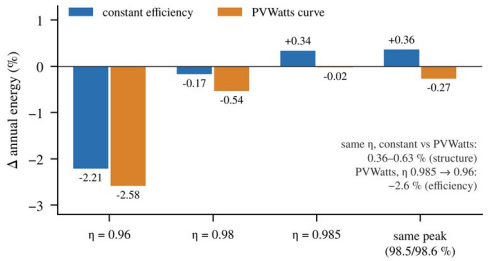


Fig. 5 Annual energy relative to PVWatts-DS for both structures at equal efficiency (as-built layout, 2023).

3.3 Weather source

Station and ERA5 irradiation differ by +2.6% mean bias and 37% RMSE (r = 0.83). The annual energy changes by +3.4%, but the monthly energy by −12% to +19% (Fig. 6), so at monthly scale the weather source outweighs the inverter model.

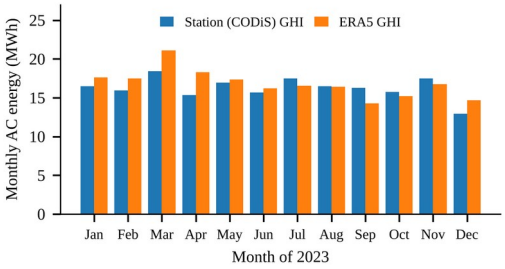


Fig. 6 Monthly AC energy driven by station or ERA5 irradiation (2023).

4. Field Validation

4.1 Measured output

On 17 and 18 October, with 97% and 94% of clear-sky irradiation at the station, respectively, the plant produced 585 and 557 kWh (4.20 and 3.99 kWh/kWp), with a peak of 100.5 kW against the 130 kW rating (Fig. 7). The one-minute power still shows some dips (variability 0.06 and 0.13 of the mean between 10:00 and 14:00, against 0.003 on a clear February day), which the AC data alone cannot explain.

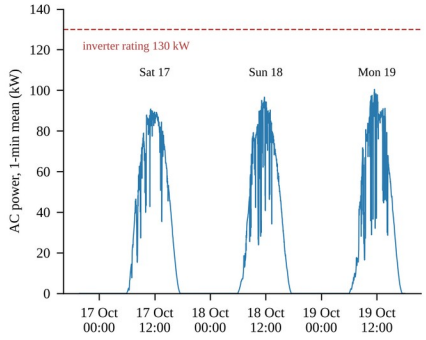


Fig. 7 Measured AC power, 16–19 October 2026 (one-minute means).

4.2 Model against measurement

Agreement is expressed by the ratio

$$R = \Sigma P_{\text{actual}} / \Sigma P_{\text{model}} \quad (4)$$

over the hours of a record ($R < 1$: model too high) and by the mean bias and RMS errors normalized by the mean measured power (Table 3). A high correlation shows only that the hourly shape is reproduced (r = 0.99 and 0.97, centroid shift 0.05 and 0.10 h); the model is higher in almost every hour, by 12.3 kW on average (Fig. 8): R = 0.81 with nMBE +23% and +24% on 17 and 18 October, and 0.80 on 19 October; the performance ratio is 0.63 measured against 0.78 modeled. The hourly ratio stays near 0.8 from morning to afternoon (morning/afternoon 0.82/0.80 and 0.76/0.86; Fig. 9), so a one-sided obstruction is unlikely. With this bias the model does not predict the hourly power on clear days.

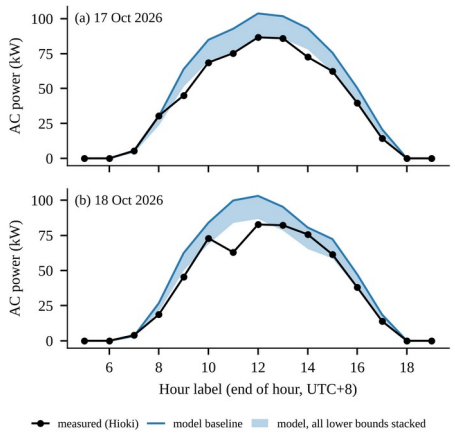


Fig. 8 Hourly AC power, measured and modeled; the band reaches the stacked lower bounds of Section 4.4.

Table 3 Error metrics on clear-sky records (hours ≥ 20 min, weighted; r for full days).

Record	h	R	MBE (kW)	nMBE (%)	RMSE (kW)	nRMSE (%)	r
17 Oct 26	11	0.81	+12.3	+23	14.2	26.7	0.99
18 Oct 26	11	0.80	+12.3	+24	15.6	30.8	0.97
10 Feb 26	3	0.89	+11.9	+13	12.0	12.7	–
21 Dec 25	2	0.82	+16.9	+22	17.5	23.2	–
7 Aug 25	3	1.01	−0.9	−1	11.2	14.6	–

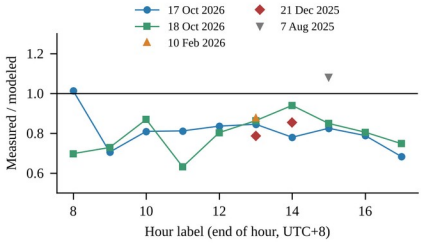


Fig. 9 Measured over modeled hourly power on the clear-sky records.

4.3 Other records

Over the nine days with station data, R ranges from 0.65 to 1.01 (Fig. 10). Only the five days above 0.9 of clear-sky irradiation are used for the bias, as a station 21 km away does not track the cloud movement over the roof; with so few records, no seasonal pattern can be separated from day-to-day differences. A single correction factor does not work either: the factor derived on 17 October (0.812) leaves a bias of +0.9% on 18 October and −0.5% on 21 December, but −8.6% on 10 February and −19.7% on 7 August. R therefore cannot be used as an annual correction.

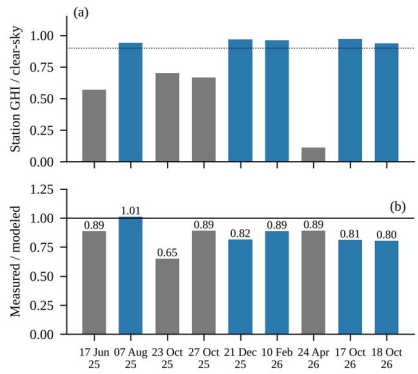


Fig. 10 (a) Clear-sky fraction of the station irradiation and (b) R per record; blue: fraction above 0.9.

4.4 Closing the budget

Fig. 11 stacks the lowest end of every factor. With isotropic transposition (−3.9% alone) and Boland decomposition (−1.0%), which the earlier stack had omitted, plus PVWatts-G, a DC loss of 11.4%, NOCT 48 °C and the incidence-angle loss, the model comes down to 87.0%; with an irradiation 5% lower it reaches 82.0%, very close to the measured 81% (R = 0.99 and 0.98). This joint extreme is implausible: the isotropic model was the

least accurate of seven transposition models in an empirical validation [27] and ignores the circumsolar brightening that dominates on clear days, the combination lowers the annual energy by 13.6% against a 5th percentile of −5.5%, and on 10 February the same setting underestimates the measurement by 6%. No single set of factor values fits the clear days of October and February; the stacked bounds are also not independent of each other.

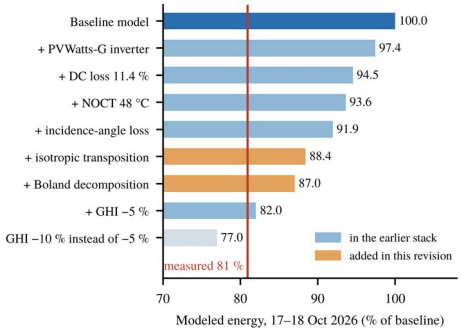


Fig. 11 Modeled energy of 17–18 October with the lower ends stacked; orange: added in this revision; red: measured.

4.5 Long-term check

The lifetime counters of the two 50 kW inverters (269,093 and 264,693 kWh after 1280 days) exceed the model for the same calendar period by 0.8% and 2.7% (1411 and 1437 against 1399 kWh/kWp per year; Fig. 12). The counter of the 30 kW inverter shows 3.6 GWh after 497,827 h, which is impossible, and is not used. The counters are the inverters' own meters (3% above the recorder on 18 October), the weather of 2024–2026 is not modeled, degradation is ignored and outages are included. Even so, a permanent 19% loss would have shown in 3.5 years, so the clear-day bias is condition-specific: either the model is lower than the plant under other conditions, or the October days are atypical. The short records, mostly from autumn and winter, cannot show which; the only summer record (7 August 2025) gives 1.01.

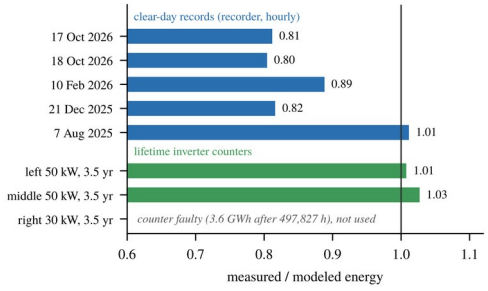


Fig. 12 Measured over modeled energy: clear-day records (recorder) and 3.5-year inverter counters (the 30 kW counter is faulty).

4.6 Causes and the inverter model

The data narrow down the cause but do not provide a definitive answer. A large bias in the station irradiation on cloud-free days is unlikely, and a fixed one-sided shadow is not supported. On the plant side, shading from the tower and frame, soiling before the rainy season, string mismatch (the 21-module string on the 30 kW inverter), de-rating of the middle inverter and short outages are all possible; on the measurement side, the uncalibrated recorder is a candidate. Separating them requires plane-of-array irradiance, inverter-level AC logging and string currents. The inverter formulations move the energy by at most 2.6% (roughly one-seventh of the gap) and give the same correlation, so the AC data alone cannot tell which is better.

5. Discussion and Limitations

The model is suitable for annual energy, within the budget of −5.5% to +11.4%, but not for hourly prediction on clear days, where it is about 20% too high. (1) The field data are three days and seven two-hour records, mostly autumn and winter, and two uncalibrated counters; utility meter data and a year of logging would allow a seasonal check. (2) Without irradiance on the roof the error cannot be split into an irradiance and a plant part; this is the main gap. (3) Only the combined AC output was measured. (4) The ranking depends on the assigned ranges. (5) The cause of the clear-day shortfall has not been confirmed.

6. Conclusions

The datasheet model gives 195 MWh (1399 kWh/kWp) per year. DC loss and irradiation error are the dominant contributors to the converged Sobol budget. The inverter effect (−2.6% to +0.4%) is an efficiency effect, not a structural one. The weather source is important at the monthly level (−12% to +19%). On clear days the plant delivered 80–89% of the modeled energy, which the budget reaches only at its joint extreme. Over the 3.5-year period the inverter counters agree with the model to within 3%. Roof-level irradiance is the next step before the inverter model is refined further.

7. Data Availability

Code and data: https://github.com/cliff1221/ncue-pv-inverter.

8. References

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